

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and applied quantitative researcher with experience developing and validating machine-learning and statistical models for financial risk, insurance pricing, and large, noisy dependent data. Builds reproducible Python/PyTorch workflows spanning data quality, feature processing, model training, simulation, diagnostics, and monitoring, and translates ambiguous problems into testable model specifications and clear recommendations.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected Jan 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Platforms: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), SQL, C++/pybind11 integration, Linux, Git, AWS EC2, HPC/SLURM; CPU/GPU computing and restartable analytical pipelines

Machine Learning and Statistics: Supervised learning, GLM/GBM, regression, feature processing and attribution, model selection and validation, hypothesis testing, Monte Carlo simulation, stochastic processes, data-quality diagnostics, and numerical optimization

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review

Summer 2026

- Developed and validated a Sequential Probability Ratio Test for binary risk indicators, translating a monitoring requirement into explicit hypotheses, likelihood-ratio boundaries, and controlled stopping decisions.
- Combined Brownian-motion approximation, dynamic-programming probability propagation, and simulation to evaluate boundary-crossing probabilities, stopping behavior, Type I/II error, and model limitations.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and performance diagnostics on AWS EC2.
- Analyzed model performance and feature-attribution patterns and presented pricing and risk-segmentation recommendations to business stakeholders.

APPLIED RESEARCH AND ENGINEERING

Ph.D. Dissertation Research - Scalable Model Development, Data Quality, and Diagnostics

Sep 2024 - Present

- **Scalable Modeling:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, replacing dense covariance operations with ordered local conditioning to fit up to 145,008 observations per day.
- **Data Quality and Root-Cause Analysis:** Built workflows for quality filtering, missingness tracking, time-dependent coordinate offsets, nearest-neighbor center matching, and regular-grid construction to separate systematic measurement artifacts from statistical model failures.
- **Model Evaluation and Monitoring:** Designed scale- and frequency-resolved diagnostics to localize whether misspecification arises from low-frequency structure, high-frequency behavior, noise, or dependence.
- **Research Engineering:** Implemented reusable Python/PyTorch pipelines with CPU/GPU execution, cached artifacts, and restartable HPC jobs; integrated compiled C++ ordering routines through pybind11.

ADDITIONAL

Coursework: Machine Learning, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Linear Algebra, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)